Week #9: [Hip Joint]

(Non-hyperlinked = Dr. Caitlin Cassidy Module)

- Introduction to Module

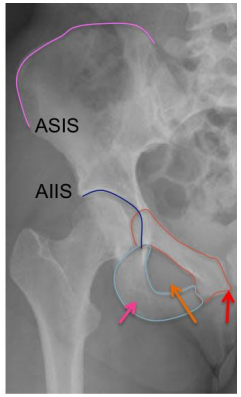
Osteology and Joints of the Proximal Lower Extremity

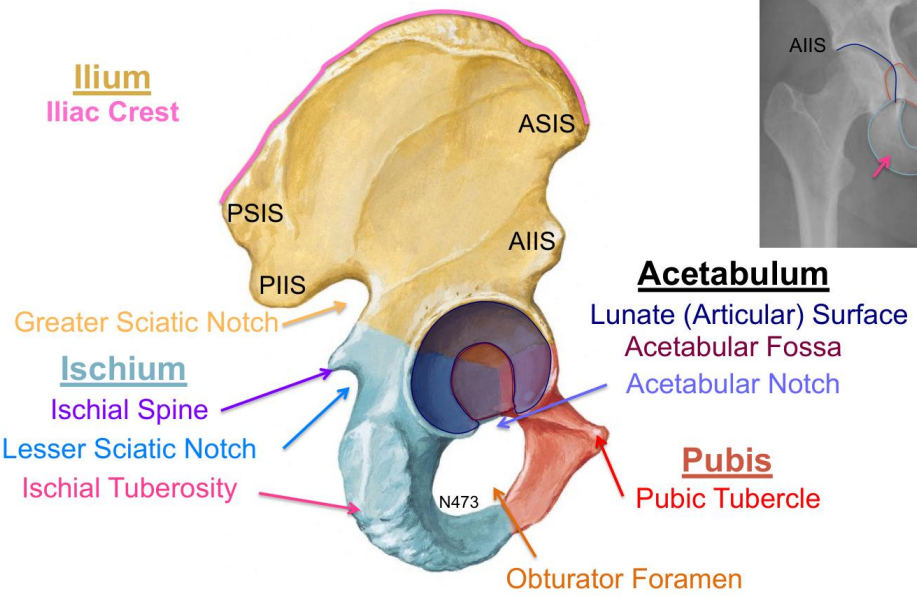
- Osteology and Joint Features of the Hip (30 min)
- Hip Fractures (26 min)



Osteology and Joint Features of the Hip

Osteology: Pelvis





I. Osteology:

a. **Hip Bone** – comprised of three bones: ilium, ischium, and pubis
   i. **Acetabulum** –articulates with the head of the femur to form the hip joint; all 3 bones (ilium, ischium, and pubis) contribute to the acetabulum

b. **Obturator Foramen** – large oval opening inferior to acetabulum bounded by the ischium and pubis; covered by the obturator membrane; allows passage of the obturator nerves and vessels

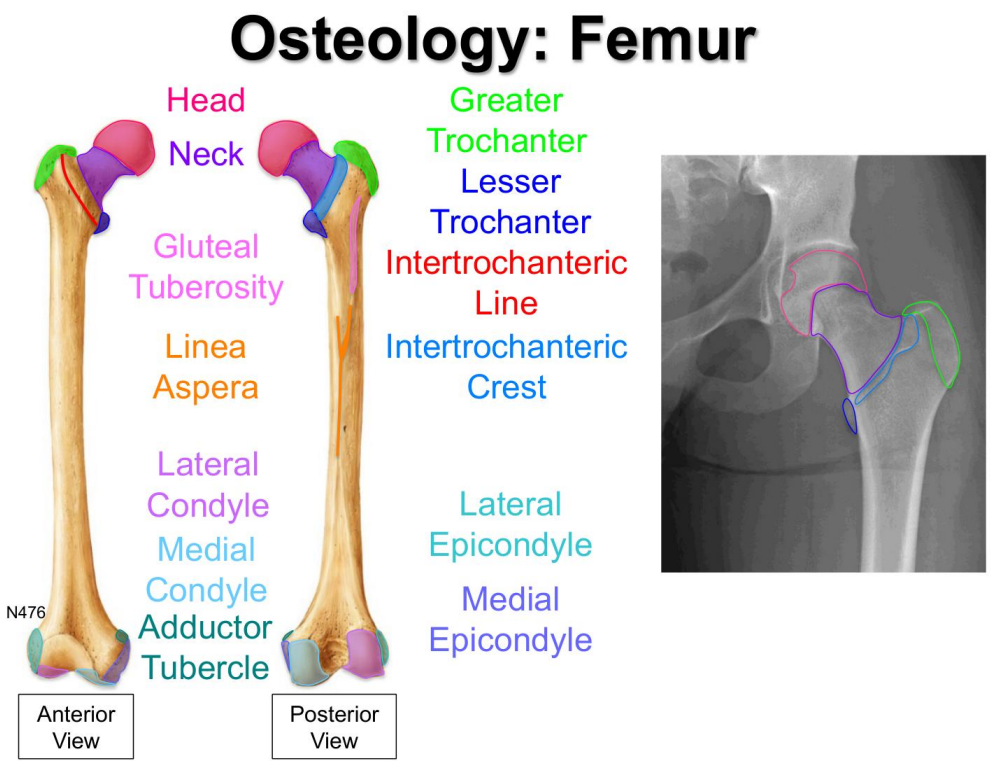
c. **Ilium** – largest part of the hip bone; forms the superior part of the acetabulum
   i. **Anterior, Posterior, and Inferior Gluteal Lines** – curved lines on lateral ilium; provide boundaries of attachment for gluteus medius and gluteus minimus
   ii. **Iliac Crest** – superior border of ilium; posterior part serves as the origin for gluteus maximus. Top of the crest corresponds to the L4 vertebral level, which can be used as a reference point when performing a lumbar puncture.
      1. **Anterior Superior Iliac Spine (ASIS)** – anterior protrusion of iliac crest
      2. **Anterior Inferior Iliac Spine (AIIS)** – inferior to the ASIS
      3. **Posterior Superior Iliac Spine (PSIS)** – posterior protrusion of iliac crest
      4. **Posterior Inferior Iliac Spine (PIIS)** –inferior to the PSIS; forms the superior border of the greater sciatic notch
      5. **Greater Sciatic Notch** – indentation on ilium that is converted to the greater sciatic foramen by the sacrotuberous and sacrospinous ligaments; enables passage of muscles and neurovasculature from the pelvis to the gluteal region

d. **Ischium** –posteroinferior portion of the hip bone; posterior part of the acetabulum
   i. **Ischial Spine** – bony spur on the posterior surface of the ischium; attachment for sacrospinous ligament
   ii. **Ischial Tuberosity** –posteroinferior protrusion of ischium; weight is placed on the ischial tuberosity when seated; origin of hamstrings
   iii. **Lesser Sciatic Notch** – indentation between the ischial spine & ischial tuberosity that is converted to the lesser sciatic foramen by the sacrotuberous and sacrospinous ligaments; enables neurovascular structures to enter the perineum from the gluteal region

e. **Pubis** –anteromedial portion of hip bone; anterior aspect of the acetabulum

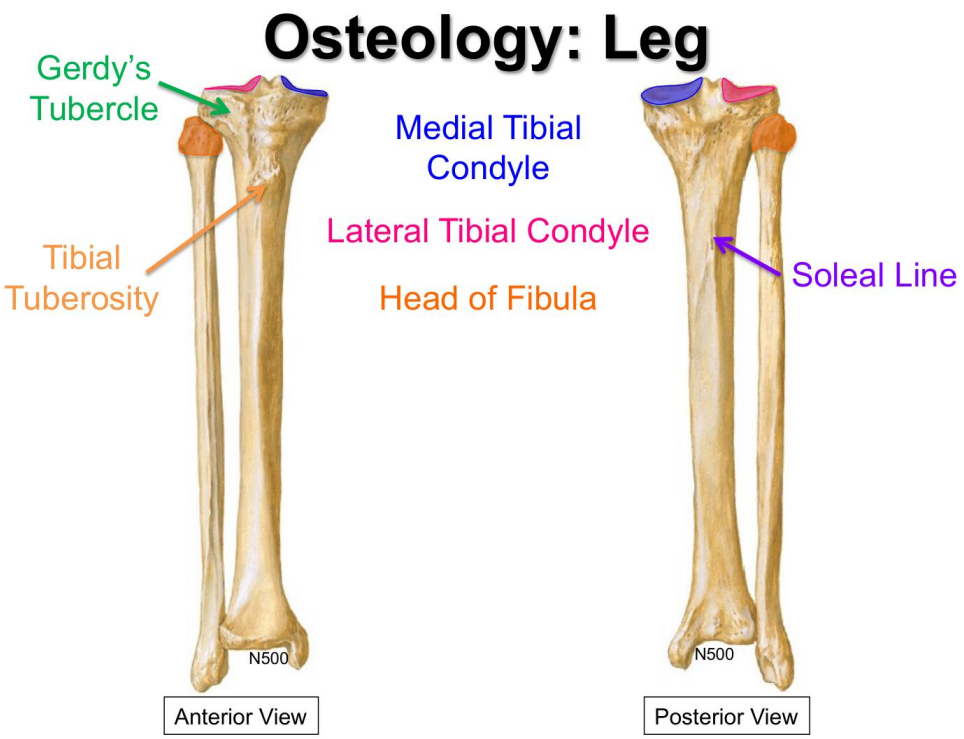
a. **Pubic Tubercle** – anterior bony protuberance; provides an attachment for the inguinal ligament
b. **Superior Ramus** – fuses with the ilium and the ischium at the acetabulum
c. **Inferior Ramus** – fuses with the ramus of ischium to form the inferior aspect of obturator foramen

F. Sacrum – articulates with the ilium to form the sacroiliac joint; serves as an attachment site for the sacrotuberous and sacrospinous ligaments and various muscles of the gluteal compartment



Femur – largest bone in the body located within the thigh; helps transmit weight from the hip to the tibia when standing. The head of the femur articulates with the acetabulum to form the hip joint.

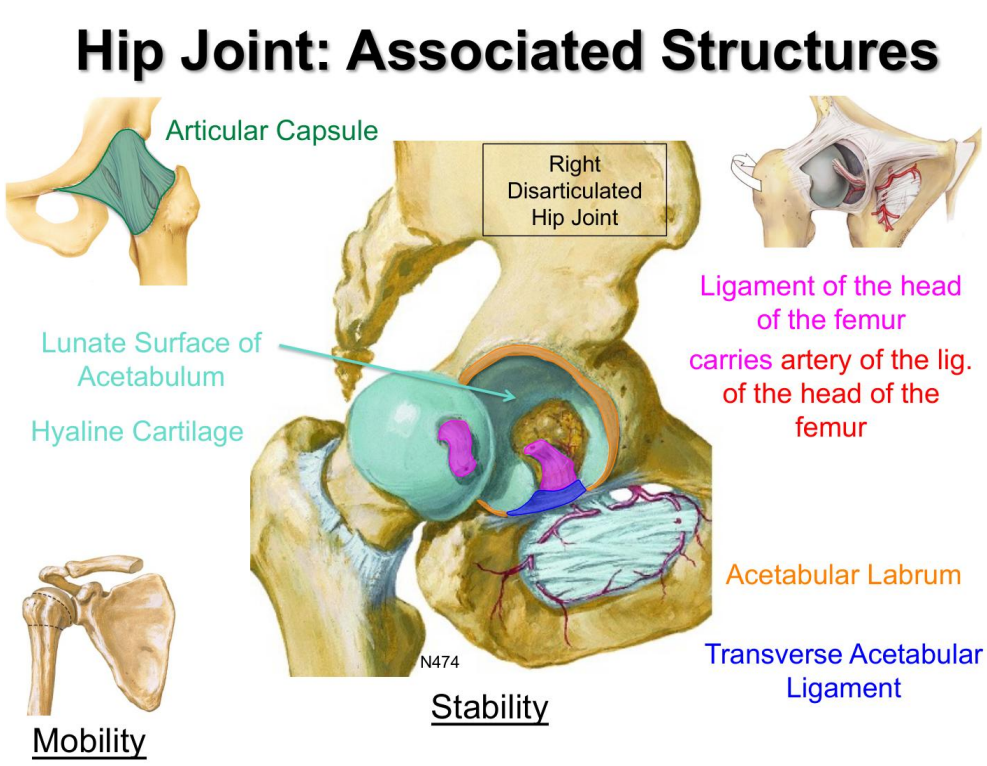
a. **Head** – proximal rounded portion; articulates with acetabulum
b. **Neck** – narrow portion connecting the head and shaft
c. **Greater Trochanter** – large superolateral bony mass at head/neck junction
d. **Lesser Trochanter** – smaller inferior posteromedial bony mass at head/neck junction
e. **Intertrochanteric Line** – anterior line between the greater and lesser trochanters
f. **Intertrochanteric Crest** – posterior ridge between greater and lesser trochanters
g. **Trochanteric fossa** – depression between the neck and greater trochanter
h. **Gluteal Tuberosity** – broad protrusion on posterior aspect of the proximal femur
i. **Linea Aspera** – rough line on the posterior surface of the shaft of the femur
j. **Medial Condyle** – medial distal end of femur; articulates with tibia
k. **Lateral Condyle** – lateral distal end of femur; articulates with tibia
l. **Intercondylar Fossa** – separates the medial and lateral condyles posteriorly



Tibia – major weight bearing bone of the leg

a. **Medial and Lateral Condyle** – articulate with femoral condyles for weight transfer
   i. **Tibial plateau** is formed by the articular surfaces of the two condyles
b. **Tibial Tuberosity** – anterior protrusion; insertion site of the patellar ligament

Fibula – the head of the fibula articulates with the tibia proximally; no weight bearing function



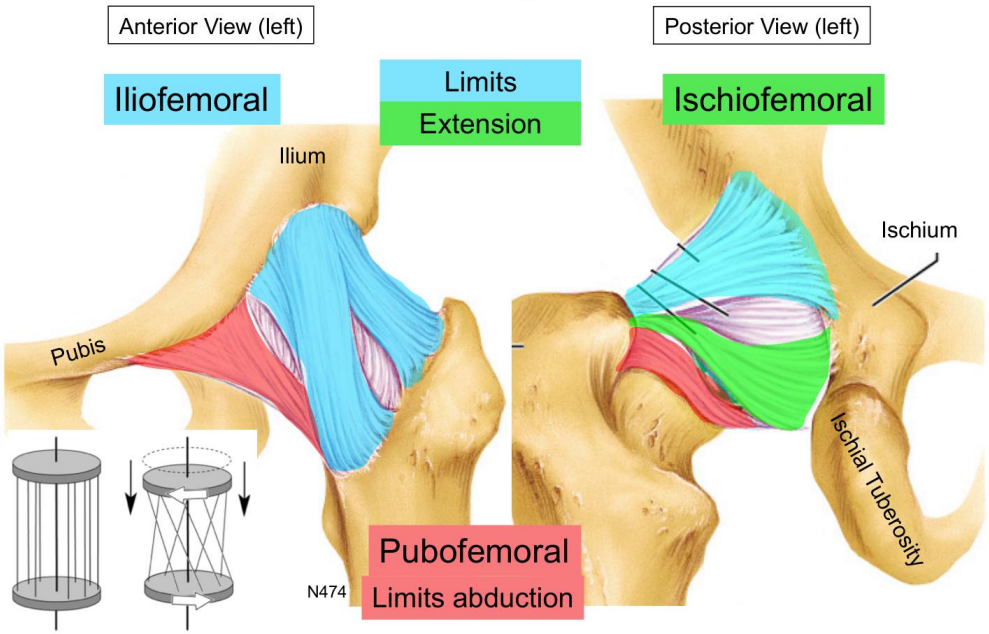
II. Hip Joint:

a. **Most stable joint** in the body; strength conveyed by:
   i. Congruency of fit (greater than 50% of femoral head fits in acetabulum)
   ii. Stout ligaments (including strongest ligament in the body)

Secondary support from extensive surrounding musculature (including dynamic support)

a. ***Type:*** ball-and-socket type of synovial joint; multiaxial
b. ***Articulation:*** acetabulum of hip articulates with head of femur
c. ***Supporting Structures:***
   i. **Joint capsule:** attaches anteriorly to **intertrochanteric line** but does NOT attach posteriorly to femur. Instead, inferior fibers of joint capsule wind posteriorly across the femoral neck (this winding adds dynamic stability to the joint, which actually 'screws' the head of the femur tightly into the acetabulum when the thigh moves into extension and medial rotation).

Hip Joint: Ligamentum



Thickenings of the capsule give rise to the ligaments of the hip joint:

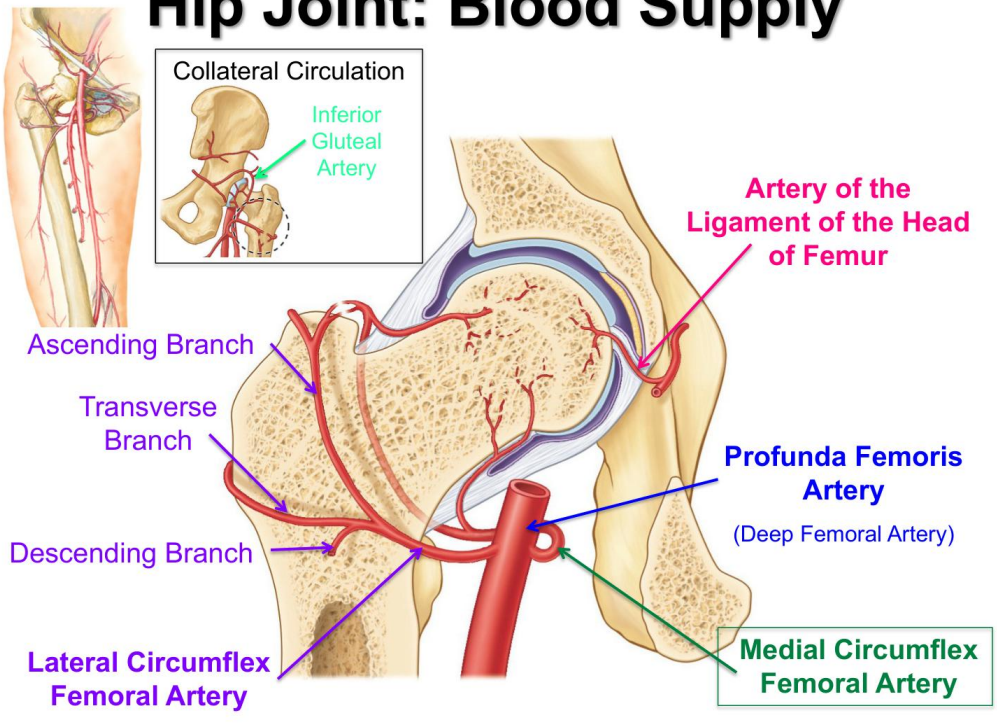
a. **Iliofemoral ligament: "Y ligament of Bigelow"**; strengthens joint anteriorly and superiorly; strongest ligament in the body
b. **Pubofemoral ligament:** strengthens joint anteriorly and inferiorly
c. **Ischiofemoral ligament:** strengthens joint posteriorly

Other structures that increase stability:

a. **Acetabular rim** gives rise to the **acetabular labrum**, a fibrocartilaginous structure that extends from the bony acetabular rim to deepen the 'socket' for the head of the femur.
b. Deficiency in the labrum at the **acetabular notch** is bridged by the **transverse acetabular ligament**

Movements: **multiaxial** and thus **very mobile**; however, the hip joint sacrifices some mobility for stability. Which is in opposition to the glenohumeral joint, which sacrifices stability for mobility.

Hip Joint: Blood Supply



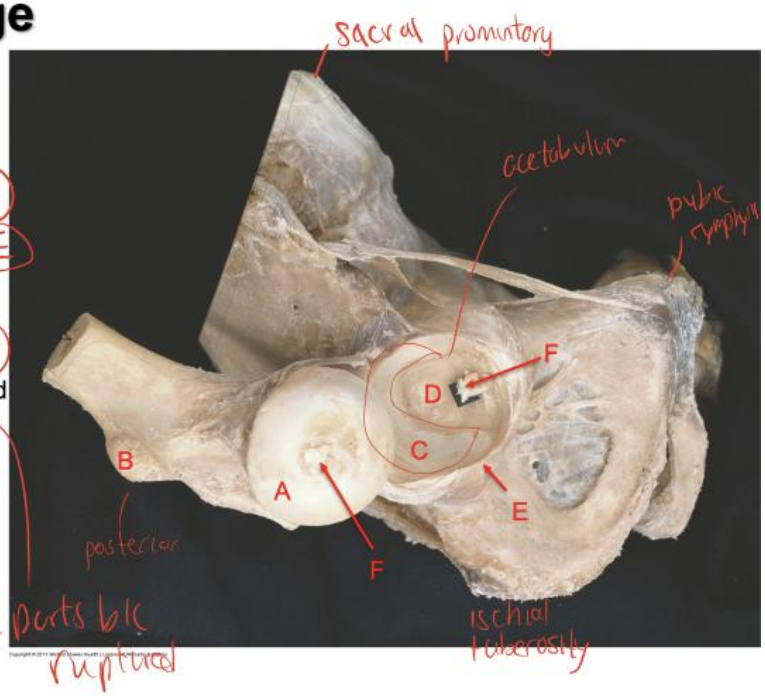
Blood supply:

a. **Medial and lateral circumflex femoral arteries** provide the majority of blood supply (***medial circumflex femoral a.***. is the most significant contributor) via retinacular arteries
b. **Artery of the ligament of the head of the femur** travels within the **ligament of the head of the femur**. The artery provides only minor blood flow to the joint.
c. **Note: ligament of the head of the femur** doesn't convey any significant strength to the joint

Clinical Note: The **inferior gluteal artery** anastomoses with branches of the medial and lateral circumflex femoral arteries providing collateral circulation around the hip joint. If there were an occlusion in the proximal femoral artery blood would still be able to supply the lower limb through this anastomosis

Clinical Note: fractures of the neck of the femur can disrupt blood supply from the circumflex arteries resulting in the artery to the ligament of the femoral head being the only remaining blood supply. This artery is frequently inadequate for maintaining the femoral head leading to **avascular necrosis of the femoral head**, likely requiring a total hip replacement. These fractures are especially common in women older than 60 years with osteoporosis

Knowledge Check:



Clinical Note: **Congenital dislocation** of the hip occurs due to the malformation of the acetabulum and/or femur at birth. This bony malformation results in less congruency at the hip joint and reduced stability. **Acquired dislocation** is an uncommon injury to the strength of the hip joint. Posterior dislocation may occur during a car accident resulting in a force being transmitted along the long axis of the femur. Patients with posterior dislocations present with the affected limb medially rotated and appear shorter than the unaffected limb. Because of the close relationship of the sciatic nerve to the hip joint it may become stretched or compressed during dislocation of the hip.

Clinical Note: **Osteoarthritis of the hip** is a common condition affecting the aging population. Osteoarthritis is associated with pathological changes within the joint affecting the articular cartilage, underlying bone, synovium, as well as the surrounding soft tissues. The degenerative pathological changes associated with osteoarthritis can result in a thickened subchondral bone plate, osteophytes, and deformation of the bony morphology. Most conservative treatments for osteoarthritis do not have long-term success; thus, severe osteoarthritis is corrected with a **surgical hip replacement.**

Hip Fractures (Dr. Cassandra Stiller-Moldovan)

- Traumatic vs Atraumatic (stress fracture)
- Epidemiology
- Risk Factors
- Clinical Presentation
- Physical Exam
- Investigations
- Fracture types
- Management
- Prognosis

Outline

- What's the big deal with hip fractures?
- Who fractures their hips?
  - Can we prevent them from fracturing their hips?
- How does a hip fracture present?
- Clinical approach to suspected/confirmed hip fracture

A Very Brief Aside...

- "Hip fracture" actually refers to proximal femur fractures. Just thought we should clear that up.

What's the Big Deal?



Source: Osteoporosis and related fractures in Canada: Report from the Canadian Chronic Disease Surveillance System 2020

Who fractures their hips? (Risk Factors)

- TAKE NOTE: This is essentially a list of things that increase your risk of FALLS (which makes you more likely to break bones) plus things that increase your risk of OSTEOPOROSIS (which makes you more likely to break bones when you fall)

- Intrinsic factors
  - Previous falls
  - Comorbidities
    - Peripheral vascular disease
    - COPD
    - Stroke
    - Other causes of weakness
  - Polypharmacy
    - Sedatives
    - Diuretics
    - Psychotropics
    - Antiarrhythmics
  - Poor nutrition
  - Poor eyesight
- Extrinsic/environmental factors

Table 1. Indications for measuring bone mineral density (BMD) [1]

Younger adults (< 50 years)
- Fragility fracture
- Prolonged use of glucocorticoids or other high-risk medication
- Hypogonadism or premature menopause (< 45 years)
- Malabsorption states or conditions
- Primary hyperparathyroidism
- Other disorders strongly associated with rapid bone loss and/or fracture

Older adults (50–64 years)
- Fragility fracture
- Prolonged use of glucocorticoids or other high-risk medication
- Parental hip fracture
- Vertebral fracture or osteopenia identified on radiography
- Excessive alcohol intake
- Smoking
- Low body weight (< 60 kg) or major weight loss (> 10% of body weight at age 25)
- Rheumatoid arthritis
- Other disorders strongly associated with osteoporosis

Seniors (≥ 65 years)
- All men and women

Hip Fracture Presentation

- History
  - O – sudden, after a fall/trauma
  - L – groin or lateral upper thigh
  - D – since fall
  - C – sharp or deep ache
  - A
    - Aggravated – movement, weight bearing
    - Alleviated – staying still
    - Associated - nil
  - R – possibly to knee/leg
  - S – ranges dramatically
- ***high suspicion in the elderly even if pain is minimal or trauma was minimal***

- Physical
  - Look
    - Limb length discrepancy (shortened), externally rotated and abducted
    - Unable to weight bear
  - Feel
    - Tenderness over hip
  - Move
    - Painful ROM, log roll, straight leg raise



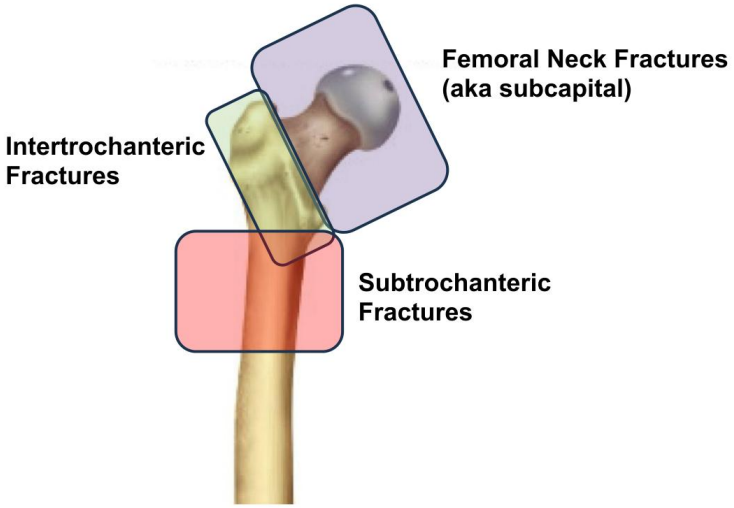
Investigations

- X-ray – AP pelvis and hip

***if X-ray negative but suspicion of hip fracture remains high, consider CT



Hip Fracture Types



Approach to Suspected Hip Fracture

1. Is this a hip fracture?
2. Is the patient stable? Are there any other associated injuries?
   - Stability is especially important to assess in a traumatic fracture in someone young (high velocity, high trauma injuries) - ABCs
   - Is there a pelvic fracture? Vascular injury? Nerve injury?
     - Check distal pulses and nerve function (especially sciatic nerve)

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3. Why did the patient have a hip fracture?

Why did they fall?

- Expand your history now that you know patient is stable!
  - –Recent illness? (nutritional factors, dehydration)
  - –Syncopal/presyncopal symptoms?
  - –Cardiac history or symptoms?

- Investigations to consider...
  - –Bloodwork
    - CBC
    - Electrolytes
    - Troponin
  - –ECG - ?atrial fibrillation or something else?
  - –Chest Xray
    - Pneumonia

Why did they fracture when they fell?

- Look on the X-ray
  - –Any signs this is a "pathologic fracture"?
    - Fracture with less trauma than should usually be required because the bone is unhealthy to begin with (ie. Bone tumour)
- Consider whether there may be a diagnosis of osteoporosis (if this is considered a fragility fracture, they automatically qualify!)
  - –Initiate osteoporosis work up as appropriate

Approach to Suspected Hip Fracture

1. Is this a hip fracture?
2. Is the patient stable? Are there any other associated injuries?
3. Why did the patient have a hip fracture?
4. What's our plan to address the fracture?

Hip Fracture Management 101

- Stabilize patient
- Decide on conservative or operative management
  - –VAST majority of all comers with hip fracture will get operative treatment because it allows a quicker return to weight bearing and reduced risk of complications from prolonged immobility
  - –Specifically for femoral neck fractures that are displaced, operative management is required to preserve the all important blood supply of the femoral head...need a reminder?

Fracture Management 101

- Stabilize patient
- Decide on conservative or operative management
  - –VAST majority will get operative treatment because it allows a quicker return to weight bearing and reduced risk of complications due to prolonged immobility
  - –Operative management also preserves the all important blood supply of the femoral neck
- Reason to consider non-surgical?
  - –Very non-displaced femoral neck fracture in a patient who is not a great surgical candidate

Operative Hip Fracture Management

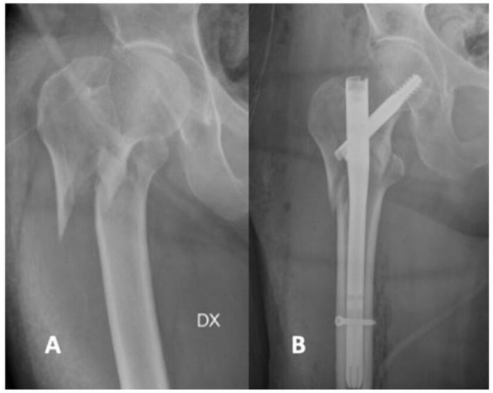
- Standard of care is surgical fixation within 24-48 hours
- Surgical approach differs based on fracture type (femoral neck, intertrochanteric or subtrochanteric) and patient factors

Femoral Neck Fractures

- Non-displaced
  - Maaaaaybe non-op
  - Maaaaaybe screws across fracture site (internal fixation)
  - Arthroplasty (hemi or total)
    - "Arthroplasty" = "arthro" (joint) + "plasty" (formation) = joint replacement
    - Hemiarthroplasty = replace one side of the joint
      - In this case, the femoral head
    - Total arthroplasty = replace both sides of the joint
      - In this case, the femoral head + acetabulum
    - Which one is selected is up to the surgeon!
- Displaced
  - Arthroplasty (hemi or total)

Other (ie. Not Femoral Neck) Hip Fractures

- Intertrochanteric and subtrochanteric fractures are usually treated the same (and essentially all are treated surgically)





Approach to Suspected Hip Fracture

1. Is this a hip fracture?
2. Is the patient stable? Are there any other associated injuries?
3. Why did the patient have a hip fracture?
4. What's our plan to address the fracture?
5. So the surgery is done, what now?
   - Post-operative considerations and complication prevention
   - Regain function

Prevent Complications

Complications are all in some way related to immobility...

- VTE is common post hip fracture
  - Best practice guidelines recommend prophylaxis (Thrombosis Canada recommends LMWH x 14-30 days)
- Pneumonia is a common cause of death post hip fracture
- Muscle atrophy
- Post-operative infection (ok maybe this is more directly related to the surgery...but UTI risk does go up with immobility!)
- Mental deterioration following surgery in older patients (high risk time for delirium)
- Bedsores from lying in the same position with minimal movement

Post-Operative Plans

- Early mobilization is important!
  - Best practice guidelines recommend interdisciplinary rehabilitation in an inpatient or outpatient setting
  - This is our best strategy in prevention of post-op complications!
- Falls prevention – don't let this happen again!
  - Optimize comorbidities
  - Consider any extrinsic factors that contributed to fall (environmental factors)
  - Involve OT/PT to reduce risk and instruct with gait aids
- Fracture prevention – don't let this happen again!
  - What about osteoporosis? Look at vit D, calcium, work up for secondary OP and consider antiresorptive therapy